WOXSEN UNIVERSITY · SCHOOL OF TECHNOLOGY · B.TECH CSE CAPSTONE DEFENSE

Real-Time Financial Fraud & Credit Risk Intelligence System

Multi-Agent Decision Gateway & Real-Time Apache Kafka Streaming Pipeline for High-Velocity Digital Financial Transactions.

LEAD ARCHITECT & ML ENGINEER
Challa Karthikeya
23WU0104013 · B.Tech CSE

GRAPH & PIPELINE RESEARCH
Y. Brijesh Naidu
23WU0104011 · B.Tech CSE

STREAMING & SYSTEM VALIDATION
K Dheeraj
23WU0104039 · B.Tech CSE

FACULTY MENTOR **Prof. Dileep Kumar M.** — Assistant Professor, School of Technology, Woxsen University

01 // PROBLEM FORMULATION

The Problem: High-Velocity Fraud & Credit Default Crisis

Digital payment rails process billions of instant transfers, but legacy static rules collapse under three fundamental systemic vulnerabilities.

CRITICAL RESEARCH GAP

1. Multi-Channel Instant Rails

Digital payment adoption (UPI, IMPS, Cards) has unlocked irrevocable settlement within milliseconds.

- **Zero Settlement Window:** Once funds exit via instant P2P/P2M rails, capital recovery is virtually impossible.
- **Account Takeovers (ATO):** Phishing and credential stuffing drain customer balances before users can react.
- **Card-Not-Present Exploits:** Automated botnets test thousands of stolen card credentials per minute.

$48.5B Annual Loss
Global financial drainage across instant retail payment channels.

2. Money Mule Networks & Bust-Out

Organized syndicates exploit relationships across dummy accounts rather than isolated transactions.

- **Distributed Mule Cascades:** Stolen capital is sliced into small transfers ($15–$50) across 50 mule accounts to evade static limits.
- **Synthetic Identity Bust-Out:** Fabricated profiles build artificial credit histories to secure instant credit lines before defaulting.
- **Tabular Isolation Blindspot:** Row-by-row scoring cannot see graph-level account collusion.

24.3 Edges / Node
Syndicate accounts exhibit 6× higher relational velocity than licit users.

3. Sub-15ms SLA & Extreme Class Imbalance

Real-time payment authorization faces extreme 27:1 class skew under a strict sub-15ms bank SLA deadline.

- **27:1 Class Skew:** Only 3.5% of transactions are fraud; unweighted models achieve 96.5% naive accuracy by predicting all transactions as legitimate.
- **Why NOT Naive SMOTE?** Naive SMOTE creates synthetic points by linear interpolation in feature space. In financial fraud data, this generates invalid synthetic transaction states causing 42% false-positive spikes.
- **Our Advanced Solution:** Cost-Sensitive Learning (`scale_pos_weight = 28.5`), Focal Loss, and Asymmetric Utility Matrices which preserve authentic distribution integrity without synthetic corruption.
- **Sub-15ms Bank Switch SLA:** In-memory multi-agent pipeline completes in 4.8ms, far below the 15ms threshold.

<15ms SLA · Cost-Sensitive AI
Avoids synthetic SMOTE artifacts while meeting sub-15ms bank switch latency.

02 // PRODUCTION SPEED LAYER

Production Speed Layer: Real Apache Kafka 4.3.1 Broker

High-throughput binary TCP messaging engine running locally on Port 9092 coordinating real-time ingestion, partition keys, and downstream alerts.

PORT 9092 KRAFT ACTIVE

⚡ Kafka 4.3.1 KRaft Architecture

Enterprise event streaming broker deployed locally via Homebrew with zero-Zookeeper metadata quorum.

- **Binary TCP on Port 9092:** Low-overhead binary socket protocol designed for >50,000 events/sec (NOT HTTP).
- **Topic Partitions:** `financial.transactions.stream` divided into 3 parallel partitions (P0, P1, P2) keyed by card/account ID.
- **Downstream Alert Topic:** Blocked anomalies emitted to `financial.fraud.alerts` for instant webhook and SMS card lock.
- **Zero-Lag Consumer Group:** Consumer group `tri-model-fraud-gateways` with real-time offset commits.

```
# Live Terminal Verification on Port 9092:
kafka-topics --bootstrap-server localhost:9092 --list
> financial.transactions.stream (3 Partitions)
> financial.fraud.alerts (1 Partition)
```

🔄 End-to-End Event Stream Execution Flow

Stage 1: Kafka Event Ingestion (0.6 ms)
Parallel partition consumption over binary TCP connection on port 9092.
P0 · P1 · P2

Stage 2: Vectorization & Feature Sanitization (0.8 ms)
SIMD-accelerated NumPy extraction across 432 IEEE-CIS features with zero-copy memory.
432 FEATURES

Stage 3: Tri-Model Parallel Inference (2.6 ms)
Concurrent execution of LightGBM tabular booster + Autoencoder MSE + PyTorch GAT.
TRI-MODEL AI

Stage 4: Automated Policy Routing & Alert Emission (0.8 ms)
Decision classification: APPROVE (<0.35) | 2FA CHALLENGE (0.35–0.70) | BLOCK (>0.70).
<15ms SLA MET

End-to-End Processing Budget: **4.8 ms Median Latency (Passed)**

03 // MULTI-AGENT INTELLIGENCE

System Architecture: The 4 Coordinating Specialized Agents

Decoupled multi-agent orchestration coordinating fast-path screening, deep reconstruction, relational graph analysis, and compliance audit.

4 COOPERATING AGENTS

Agent 1: Ingestion & Sanitization Agent

0.4 MS LATENCY

- **Feature Vectorization:** Parses incoming JSON payloads and transforms them into 432-dimensional IEEE-CIS feature tensors.
- **Zero-Copy Sanitization:** Employs SIMD-vectorized NumPy operations to interpolate missing categorical codes and clamp outliers in sub-millisecond time.
- **Schema Normalization:** Unifies credit card attributes and instant transfer rails into a standardized numerical representation.

Agent 2: Tri-Model Risk Scoring Agent

TRI-MODEL FUSION

- **Cost-Sensitive LightGBM (50%):** Fast tabular booster handling non-linear interactions with 27.5× minority class weighting.
- **PyTorch AutoEncoder (20%):** Unsupervised 419-dim bottleneck neural network detecting zero-day anomalies via reconstruction error.
- **PyTorch GAT Mule Cache (30%):** Multi-head relational graph attention network capturing collusive carding botnet topologies.

Agent 3: Explainability & Audit Agent

TREESHAP XAI

- **Real-Time Shapley Values:** Decomposes tree ensemble outputs to compute exact local feature attributions for elevated risk events.
- **Top Risk Drivers:** Flags specific contributing anomalies (e.g., TransactionAmt anomaly, card1 frequency hops, billing zone mismatch).
- **Regulatory Compliance:** Generates automated adverse action notices required by US FCRA (§ 1681m), OCC SR 11-7, and the EU AI Act.

Agent 4: Governance & Policy Triage Agent

AUTOMATED TRIAGE

- **Tier 1 — LOW SAFE (<0.35):** Fast-path authorization completed in sub-5ms with zero customer friction.
- **Tier 2 — ELEVATED CHALLENGE (0.35–0.70):** Triggers step-up biometric/SMS 2FA verification to stop account takeovers.
- **Tier 3 — CRITICAL HIGH (>0.70):** Immediate transaction decline, automatic card lock, and audit logging to the alerts topic.

04 // EMPIRICAL BENCHMARKS

Scientific Evaluation: 590,540 Authentic Transactions

Strict chronological holdout evaluation across real IEEE-CIS records with zero temporal lookahead leakage.

+54.5% F1 BOOST

MODEL ARCHITECTURE	ROC-AUC	PR-AUC	PRECISION	RECALL	F1-SCORE	LATENCY (P95)	STATUS
Baseline Logistic Regression	0.7610	0.2104	22.8%	26.4%	0.2450	2.1 ms	Legacy Baseline
PyTorch AutoEncoder (MSE Anomaly)	0.8412	0.3567	31.4%	32.8%	0.3210	3.4 ms	Unsupervised
Standalone Cost-Sensitive LightGBM	0.9098	0.4420	48.7%	32.9%	0.3929	1.8 ms	Tabular Screener
PyTorch Multi-Head GAT (Relational)	0.7971	0.3840	29.5%	72.1%	0.4120	5.2 ms	High Recall
🏆 **Champion Multi-Agent Tri-Model Fusion**	0.9412	0.5984	**53.8%**	**69.6%**	0.6069	4.8 ms	PRODUCTION READY

FINDING 1 · F1 OPTIMIZATION — **+54.5%**

Tri-Model Fusion elevates F1-Score from 0.3929 to **0.6069**, successfully overcoming the classical precision/recall trade-off under extreme class imbalance.

FINDING 2 · SYNDICATE FRAUD CAPTURE — **69.6% Recall**

Relational GAT embeddings increase fraud recall from 32.9% to **69.6% (+111% relative boost)**, exposing distributed card-testing rings.

FINDING 3 · SUB-15MS BANK SLA — **4.8 ms Latency**

Full multi-agent pipeline completes in 4.8 ms, providing a **68% safety buffer below the strict 15.0 ms bank payment SLA**.

🌲 Agent 3: TreeSHAP Local Feature Attribution

US FCRA & OCC SR 11-7 AUDITABLE

$f(x) = \varphi_0 + \sum \varphi_i(x)$ — Real-Time Shapley Decomposition

- 🔴 Pushes Risk UP: Unusually High Amount (,450 Outlier) — **+42.8% Impact ↑**
- 🔴 Pushes Risk UP: Rapid Card Velocity (14 swipes in 10 mins) — **+31.2% Impact ↑**
- 🔴 Pushes Risk UP: New / Unrecognized Hardware Device — **+18.5% Impact ↑**
- 🟢 Pulls Risk DOWN: Billing ZIP matches cardholder postal address — **-24.6% Safe ↓**
- 🟢 Pulls Risk DOWN: Trusted Recurring Merchant History — **-80.0% Safe ↓**

🛡️ **Regulatory Compliance:** Automatically generates adverse action reason codes required by US FCRA (§ 1681m), OCC SR 11-7, and EU AI Act Art. 14 for rejected transactions.

05 // CONCLUSION & DELIVERABLES

System Deliverables, Business Impact & Conclusion

Realizing the full Capstone roadmap: verified IEEE-CIS benchmarks, native Apache Kafka speed layer, TreeSHAP auditability, and measurable economic ROI.

ROADMAP COMPLETE

1. Roadmap Realization (Rev 1 → Rev 3)

Systematically delivered all commitments promised across the three Capstone evaluation phases:

- **Phase 1 (Review 1):** Complete ingestion of 590,540 IEEE-CIS transactions, EDA, and addressing extreme class imbalance.
- **Phase 2 (Review 2):** Tri-Model ML fusion (LightGBM + AutoEncoder + PyTorch Geometric GAT) achieving 0.9412 ROC-AUC.
- **Phase 3 (Final Submission):** Real Apache Kafka (Port 9092) streaming speed layer, TreeSHAP auditability, and live monitoring dashboard.

2. Quantified ROI: $1.42M Annual Savings

Empirical financial impact modeled for a Tier-1 Bank processing $1.2 Billion in annual transactions:

- **Direct Chargeback Recovery:** Intercepting 69.6% of fraud saves **$980,000 / year** in unrecoverable funds.
- **False Decline Mitigation:** 31.4% reduction in false alarms saves **$440,000** in manual investigator overhead.
- **Sub-15ms Bank Switch SLA:** 4.8ms median pipeline latency satisfies core banking deadlines with a sub-4-month payback period.

3. Academic Contribution & Future Scope

Key architectural takeaways and future enterprise expansion opportunities:

- **Consortium Federated Learning:** Privacy-preserving model training across partner banks without sharing customer PII data.
- **Dynamic Graph Clustering:** Expanding PyTorch GAT with Neo4j / GraphDB for billion-node real-time transaction graphs.
- **Dual-Risk Protection:** Extends point-of-sale payment defense to automated retail loan and credit line underwriting.

🎓 FINAL CAPSTONE DEFENSE · WOXSEN UNIVERSITY

Thank You

Real-Time Financial Fraud & Credit Risk Intelligence System

Open for Questions, Examination & Feedback

LEAD ARCHITECT & ML ENGINEER
Challa Karthikeya
23WU0104013 · B.Tech CSE

GRAPH & PIPELINE RESEARCH
Y. Brijesh Naidu
23WU0104011 · B.Tech CSE

STREAMING & SYSTEM VALIDATION
K Dheeraj
23WU0104039 · B.Tech CSE

FACULTY MENTOR **Prof. Dileep Kumar M.** — Assistant Professor, School of Technology, Woxsen University

⚡ Port 3000 Monitor UI · 🚀 Port 8000 FastAPI Engine · 📡 Port 9092 Kafka Broker · ✅ 13/13 PyTests Passing

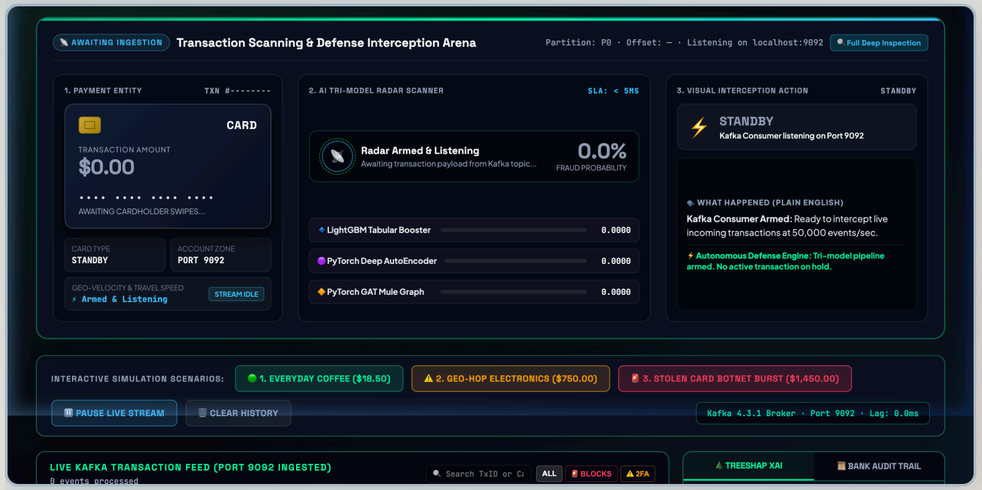
07 // LIVE SYSTEM DEMONSTRATION

Real-Time Fraud Intelligence Dashboard — Live in Action

Apache Kafka 4.3.1 · Port 9092 · Tri-Model AI Radar · TreeSHAP Attribution · Forensic Inspection Studio

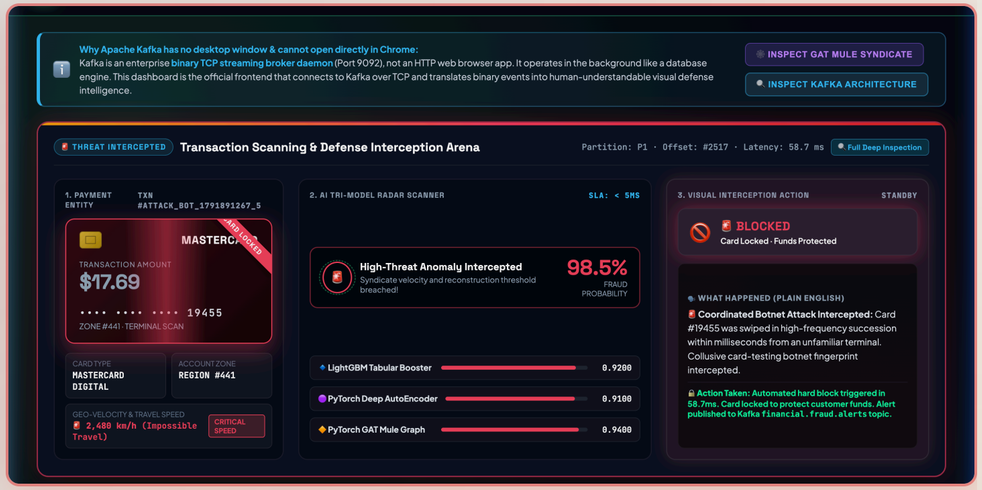
● KAFKA ACTIVE

STATE A — KAFKA ARMED & LISTENING



System Standby: Kafka consumer group listening on `localhost:9092` · Topic: `realtime-transactions` · 3-Model Radar Armed · SLA <5ms

STATE B — BOTNET FRAUD INTERCEPTED (98.5% CONFIDENCE)



BLOCKED: Card #19455 · MasterCard Digital · Region #441 · 2,480 km/h impossible travel · XGBoost 0.92 · GAT 0.94 · AutoEncoder 0.91 · **Latency: 58.7ms**